

Causal normalizing flows: from theory to practice

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Normalizing Flows

Causal Inference

Interventions & Counterfactuals

Why this matters in the real world

Fairness harms

- Credit scoring on mixed data can unfairly penalize groups if causal effects are ignored.
- Auditing needs counterfactuals: what if sex were different while all else stays causally consistent?
- Partial graphs and discrete features are common; per-variable DNNs propagate errors.

Concrete example (German Credit)

- Observed data only; estimate unseen effects of $\text{do}(x_S=s)$ on checking account and credit risk.
- Counterfactually unfair models can flip decisions solely due to sex, harming qualified applicants.
- We need a single, evaluable model to support do-queries without error compounding.

Observed vs interventional distributions for checking account

Observed (blue) vs interventional (red/yellow) for checking account under $\text{do}(x_S=0/1)$: interventional shifts reveal potential unfair influence; recovering them from observational data enables fair auditing.

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Code: <https://github.com/pasch21/causal-flows>

Fairness audit: using exogenous variables from the causal NF (excluding u_S) yields near-equal accuracy with zero counterfactual unfairness—mitigating real harms in lending.

User story and setup: German Credit

User story

- Anna and Ben have similar finances; differing only in sex (x_S). A naive model changes the loan decision when toggling x_S .
- With a causal NF, we compute counterfactuals that hold non-descendants fixed and isolate unfair influence, preventing harmful flips.
- Outcome: decisions reflect causal factors (income, history), not proxies of sex.

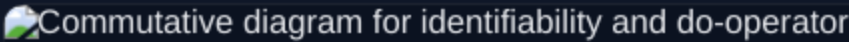
Setup

- Mixed discrete–continuous features incl. sensitive attribute sex (x_S).
- Partial graph via blocks with known relations; rest unknown.
- Goal: answer do-queries and train counterfactually fair classifiers.

Fairness results preview

Method intuition: identifiability and interventions

- Abductive ANF $T_\theta: x \rightarrow u$ with $L=1$; mask by A when known; train by MLE on $p(x)$.
- Theorem 1: two TMI maps with factorized P_u matching $p(x)$ differ by invertible per-component transforms $\rightarrow u$ isolated.
- Corollary 2: Jacobian sparsity matches SCM ($\nabla_x T_\theta \equiv I-A$), enabling faithful do-queries.

Commutative diagram for identifiability and do-operator

How to read the diagram: top path maps $u \rightarrow x$ (SCM); left path applies T_θ to isolate u ; changing P_u (green path) pins u_i for $\text{do}(x_i=\alpha)$, keeping f fixed. Implication: interventions alter only the exogenous component for x_i , preserving causal pathways.

- Observed vs $\text{do}(\cdot)$: observed mixes all causes; $\text{do}(x_i=\alpha)$ breaks incoming edges to x_i by fixing the exogenous drive for that node.
- Contrast: differences isolate causal effect of x_i on its descendants; ancestors no longer influence x_i under do .

- Practice: compute $u=T_\theta(x)$, set $x_i:=\alpha$ and u_i accordingly, then propagate to sample interventional/counterfactuals.

Crisp problem and approach

- Problem: learn a single flow for the full causal process; support interventions/counterfactuals from observational data.
- Approach: abductive causal NF ($x \rightarrow u$, $L=1$); mask by A or regularize to match $\nabla_x T_\theta$ sparsity; MLE training.
- Why it works: identifiability up to element-wise transforms; causal consistency reduces spurious paths and error propagation.

Observed vs interventional

Observed vs $\text{do}()$ contrast

Observed distribution blends causes; $\text{do}(\cdot)$ reveals pure downstream impact. Reading gap between curves isolates the causal effect.

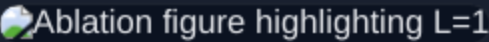
Do-operator (implementation)

Modify P_u : set u_i consistent with $\text{do}(x_i=\alpha)$; keep f unchanged. Works with mixed data and partial graphs via masking/blocks.

Operating in exogenous space ensures $x_i=\alpha$ regardless of ancestors, enabling valid interventions across architectures.

Results and takeaways

Ablation: L=1 abductive wins

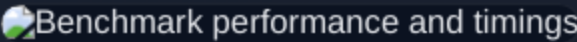
Ablation figure highlighting L=1

Single-layer abductive with graph masking is causally consistent by design, giving best ATE/CF with simplest model.

Generative needs $L \geq \text{diam}(A)$ to avoid shortcuts

L=1 (abductive, graph) → lowest ATE RMSE

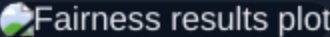
Benchmarks

Benchmark performance and timings

Across non-linear SCMs, causal NF matches/outperforms CAREFL on KL/ATE/CF while being faster; VACA is slower and less accurate.

Fairness vignette (German Credit)

Classifier	Variant	Acc (%)	CF unfair.
Logistic	full	67.0	5.84
Logistic	unaware	66.8	2.81
Logistic	fair u	66.5	0.00
SVM	full	69.5	6.65
SVM	unaware	71.0	2.78
SVM	fair u	69.8	0.00

Fairness results plot

Key takeaway: using non-sensitive exogenous variables (fair u) reduces counterfactual unfairness to 0.00 with similar accuracy to full/unaware variants—enabling fair, reliable decisions.

Simple, causally consistent design → better ATE/CF

Faster than baselines; enables fair auditing